



Zomato Delivery Performance Analysis

Factors Influencing Delivery Time



Introduction

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Zomato Delivery Performance Analysis: Factors Influencing Delivery Time

This project focuses on analyzing the key factors that influence Zomato's delivery time. External factors include traffic conditions, weather, and delivery distance, while internal factors cover driver age, vehicle condition, and multiple deliveries. The analysis shows that heavy traffic, bad weather, and longer distances increase the risk of delays, while poor vehicle conditions and multiple orders add more variance to delivery time.



Objectives

1

Analyze key factors influencing delivery performance, particularly delivery time

2

Compare delivery times across different operational conditions such as traffic density, weather, and vehicle condition

3

Provide actionable insights to improve operational efficiency and customer satisfaction



Goal

To enhance delivery efficiency and service consistency by identifying the key factors that influence delivery time, enabling data-driven decisions to reduce delays, improve on-time delivery rates, and maintain high customer satisfaction

Tools



Power BI



Streamlit

Project Background



In the highly competitive food delivery industry, delivery time is one of the most critical factors influencing customer satisfaction and loyalty. Even small delays can impact customer experience, restaurant operations, and overall service quality. With rising customer expectations and increasing competition, companies like Zomato must identify and address the key factors that affect delivery time to maintain efficiency and competitiveness.



Project Overview

This project evaluates Zomato’s delivery performance using historical operational data on drivers, locations, environmental and vehicle conditions, and orders. **Early findings reveal notable delivery time variations for similar distances, suggesting inefficiencies in route planning, driver allocation, or operations.** The analysis aims to identify the drivers of these inconsistencies and their impact on customer experience and business results

Why This Project Matters?

Consistent delivery time is crucial for customer satisfaction, efficiency, and competitiveness. Understanding the internal (driver skill, vehicle condition, workload) and external (traffic, weather, Vehicle Condition) factors behind these variations enables Zomato to implement targeted improvements and strengthen on-time performance





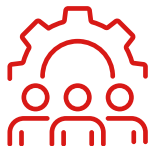
Expected Outcomes

1 Clear identification of factors with the greatest impact on delivery time

2 Comparative analysis of delivery time under different operational conditions (traffic, weather, vehicle condition)

3 Actionable insights to improve operational efficiency and customer satisfaction

Beneficiaries

 The **operations team** can optimize driver allocation and delivery routes

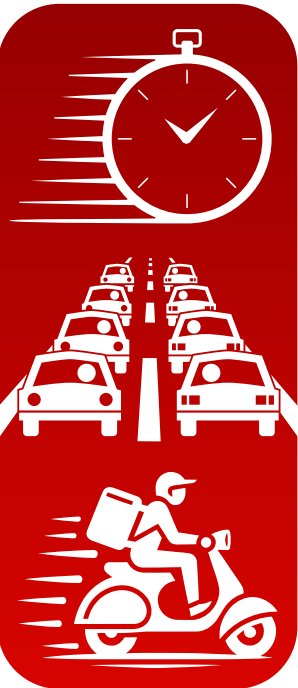
 **Management** can make well-informed operational decisions

 **Restaurant** partners benefit from more dependable delivery services

 **Customers** enjoy faster and more consistent deliveries



Problem Statement



- Zomato experiences inconsistent delivery times** that may reduce customer satisfaction and efficiency
- Factors like traffic, weather, and vehicle condition may cause delays**, but their exact impact is unclear
- Fluctuating delivery performance** makes it hard to maintain service quality and competitiveness

Why This Problem Matters

- 1 Delivery Time Drives Business Success**
Delivery time is a critical factor in the food delivery industry, directly influencing customer satisfaction, loyalty, and brand reputation
- 2 Lack of Clarity on Delay Causes**
Unclear causes of delays prevent Zomato from implementing targeted operational improvements to address the root issues
- 3 Operational and Competitive Gains from Resolution**
Resolving this problem will improve operational efficiency, increase on-time delivery rates, and strengthen Zomato's market competitiveness

Business Impact – If the Problem Is Resolved



Data Understanding & Preprocessing



Dataset from Kaggle



Zomato Delivery Operations Analytics Dataset
Exploring Delivery Dynamics and Customer Experience on Zomato
[kaggle.com](#)



45584

Number of Records

20

Number of Variables

9 Numeric

11 Categorical

The Zomato Delivery Dataset contains historical operational data, including driver information, ratings, locations, order times, distances, and environmental conditions

Number of duplicate rows : 0 Duplicates

Note : All other columns have 0 missing values

	Missing Values
Delivery_person_Age	1854
Delivery_person_Ratings	1908
Time_Orderd	1731
Weather_conditions	616
Road_traffic_density	601
multiple_deliveries	993
Festival	228
City	1200

Handling Missing Values (dropna method)

Step	Number of Records	Missing Values
Before Cleaning	45,584	Yes
After Removing	41,359	No

Removed ~9.27% of data

Data Type Conversion

Order_Date

Converted to datetime format for accurate date-based analysis

Time_Orderd

Converted to time format for better time-based calculations

Time_Order_picked

Converted to time format for better time-based calculations

No invalid values were found in the dataset

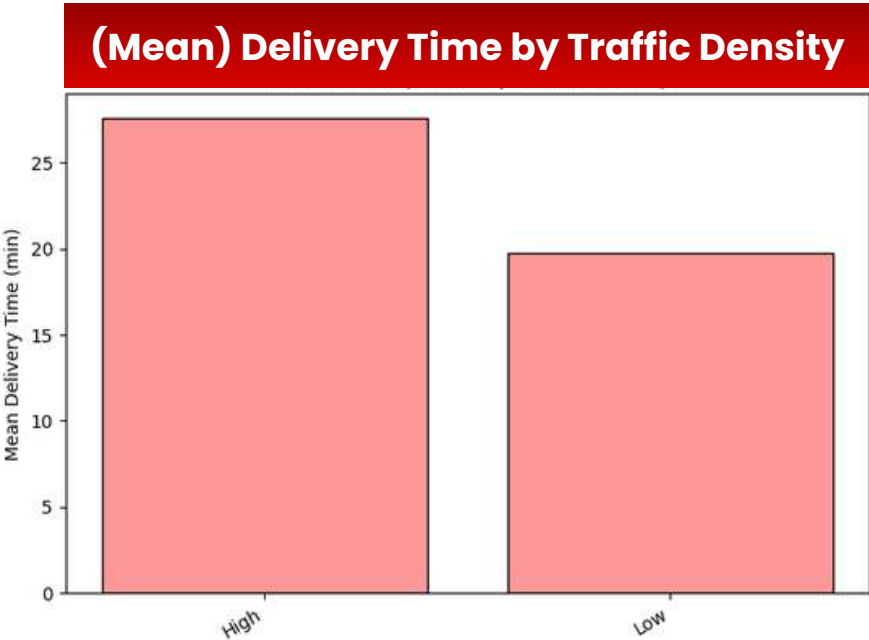


Technique Used

The analysis applied **descriptive** and **exploratory data analysis (EDA)** techniques to uncover patterns in delivery performance

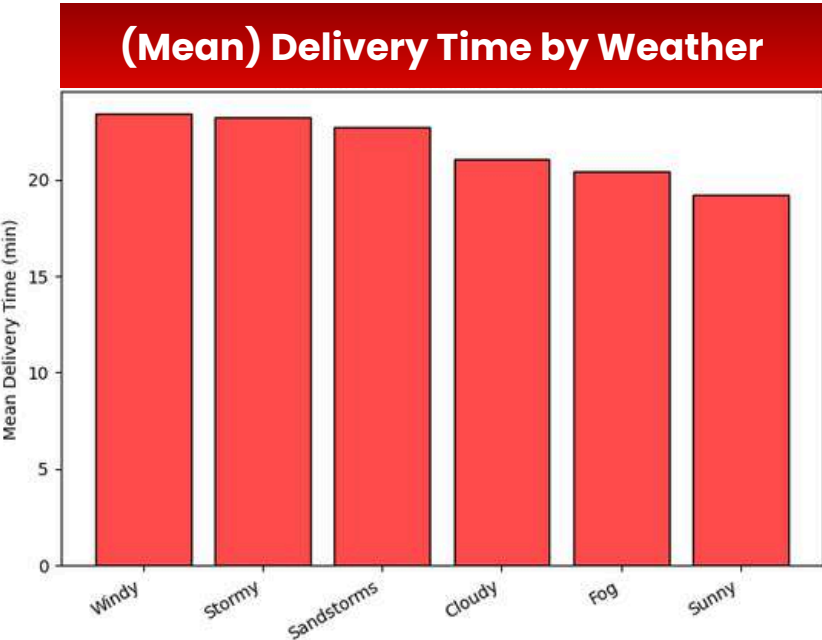
External Factors Affecting Delivery Time on Zomato

Based on the predominant delivery distance of 1.5 km (using 1-2km)



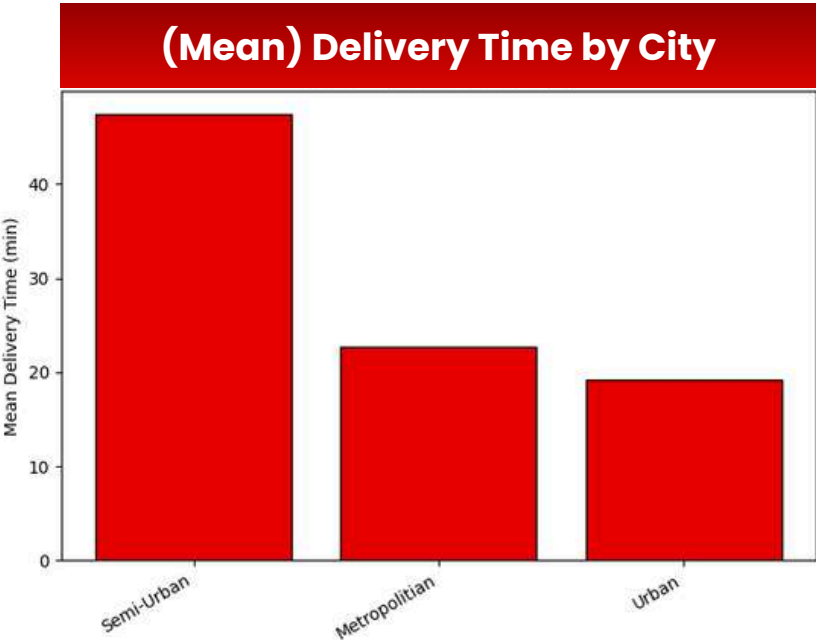
Mean Delivery Time by Traffic Density		
road_traffic_density	mean_time	
High	27.610262	
Low	19.728359	

Deliveries in **high traffic areas** take **around 27.6 minutes** on average, which is **significantly slower** compared to 19.7 minutes in low traffic, indicating congestion plays a major role in delays



Mean Delivery Time by Weather		
weather_conditions	mean_time	median_time
Windy	23.392361	24.0
Stormy	23.242284	24.0
Sandstorms	22.734597	23.0
Cloudy	21.034258	19.0
Fog	20.432343	19.0
Sunny	19.232945	18.0

Windy and stormy weather extend delivery times to about 23.3 minutes, whereas **sunny conditions are faster** at 19.2 minutes, suggesting adverse weather disrupts delivery efficiency



Mean Delivery Time by City		
city	mean_time	median_time
Semi-Urban	47.500000	47.5
Metropolitan	22.671012	22.0
Urban	19.250234	18.0

Semi-urban areas show **extremely long delivery times**, averaging 47.5 minutes, compared to 19.3 minutes in urban areas, likely due to infrastructure and accessibility challenges

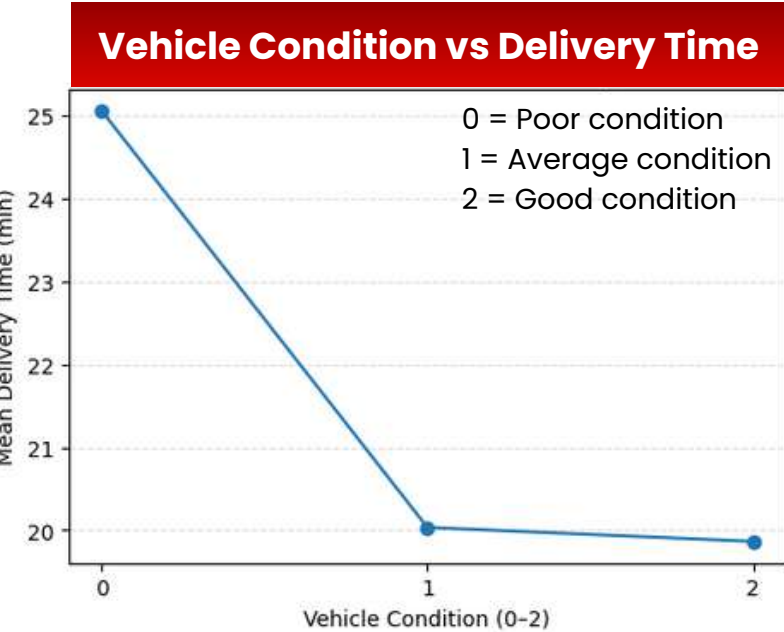


Mean Delivery Time by Festival		
festival	mean_time	median_time
Yes	44.833333	44.5
No	21.578027	21.0

During **festivals**, delivery times spike to an **average of 44.8 minutes**, more than double the 21.6 minutes on **normal days**, indicating that seasonal demand significantly slows down delivery operations



Internal Factors Affecting Delivery Time on Zomato

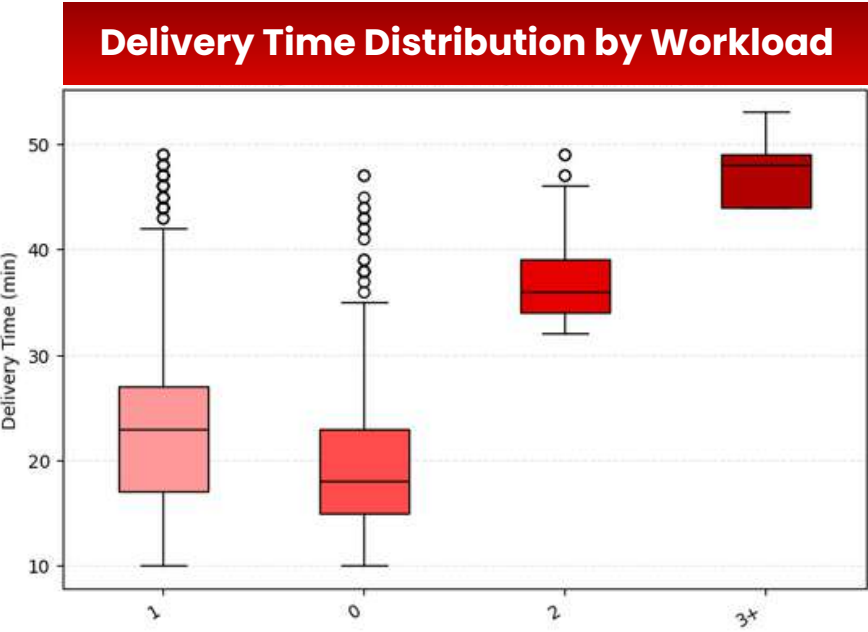


Poorly maintained vehicles (condition 0) take the longest to deliver at ~25 minutes on average, whereas well-maintained vehicles (condition 2) reduce delivery time to under 20 minutes, showing the operational impact of maintenance

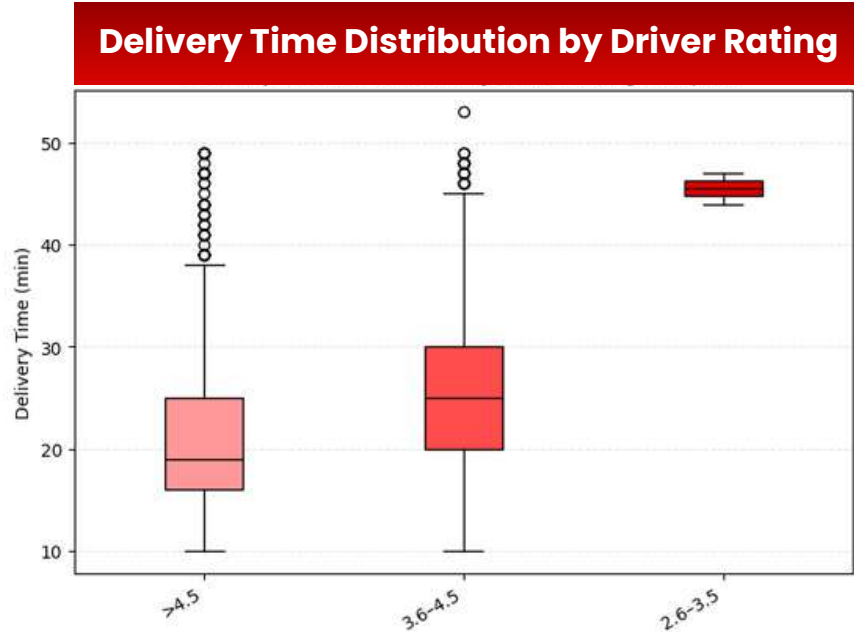


Younger drivers (<25) tend to deliver slightly faster, with a lower median time and fewer extreme delays, while older age groups (35-44) show higher median times and more variability

High workloads (3+ orders) cause a dramatic slowdown, pushing median delivery times close to 50 minutes, compared to around 17-25 minutes when handling one or zero orders



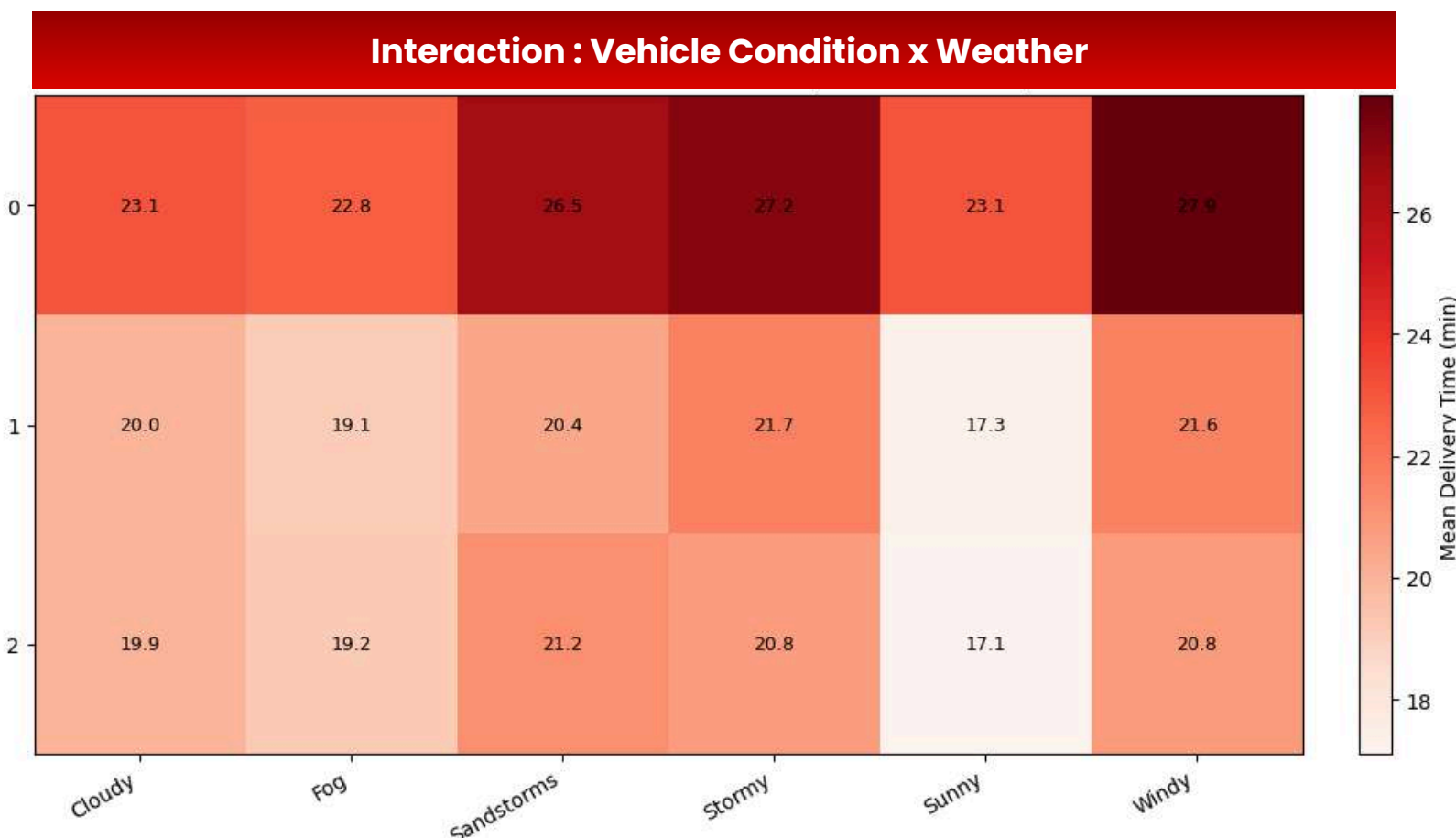
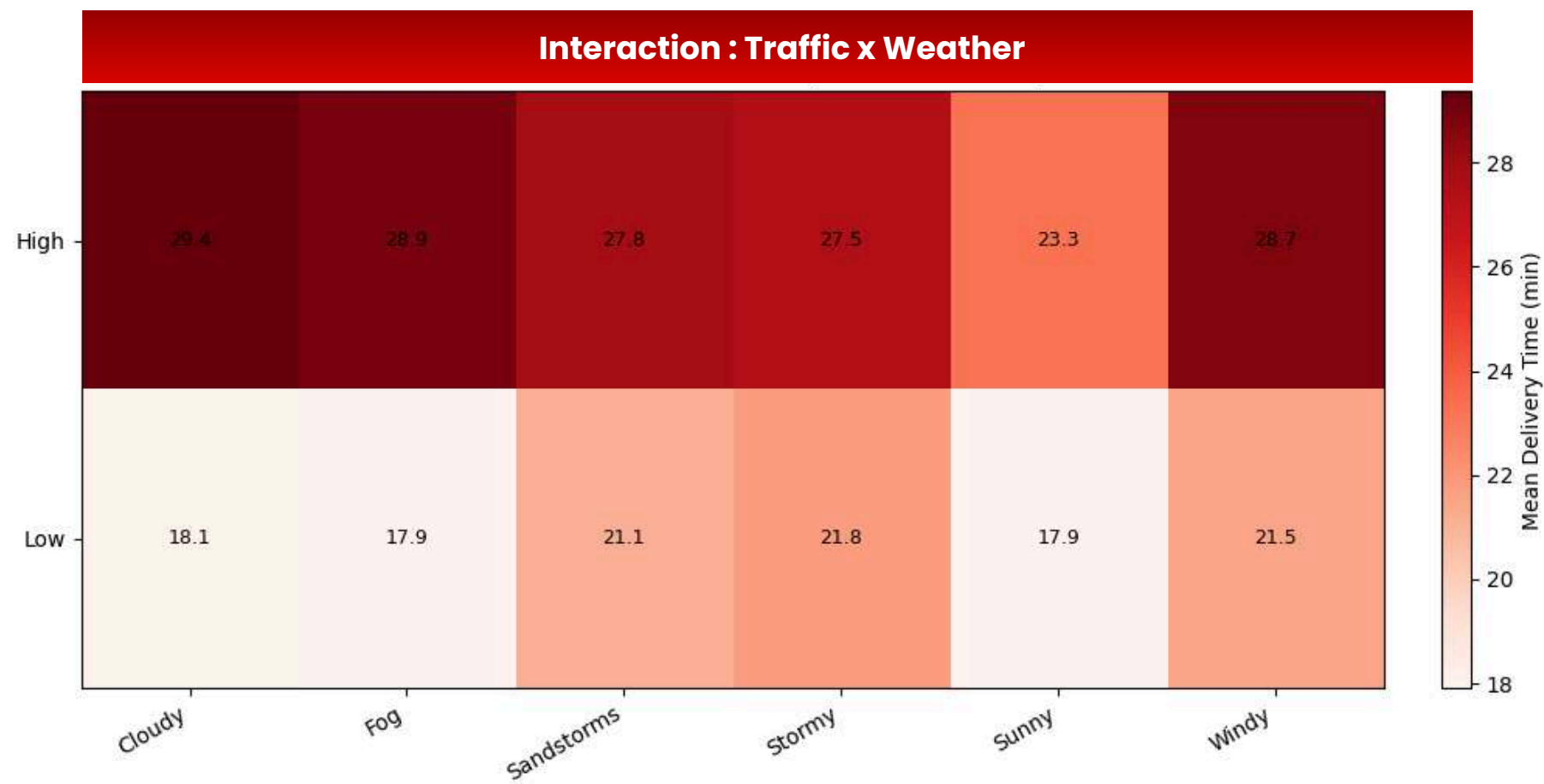
Drivers with ratings above 4.5 complete deliveries faster and more consistently, while those rated 2.6-3.5 have both the slowest delivery times (median ~46 minutes) and the least variability, possibly due to consistently poor performance



*Internal factors are variables originating within the organization's operations that can be directly managed or influenced, while external factors are variables arising from the surrounding environment that are beyond the organization's direct control



Interaction analysis



High traffic combined with cloudy, foggy, or windy conditions results in the longest delivery times, reaching around 29 minutes

Low traffic consistently leads to faster deliveries across all weather types, indicating that traffic is a more dominant factor than weather in slowing down deliveries

Poor vehicle condition (0) combined with extreme weather such as sandstorms, stormy, or windy conditions leads to the highest delivery times

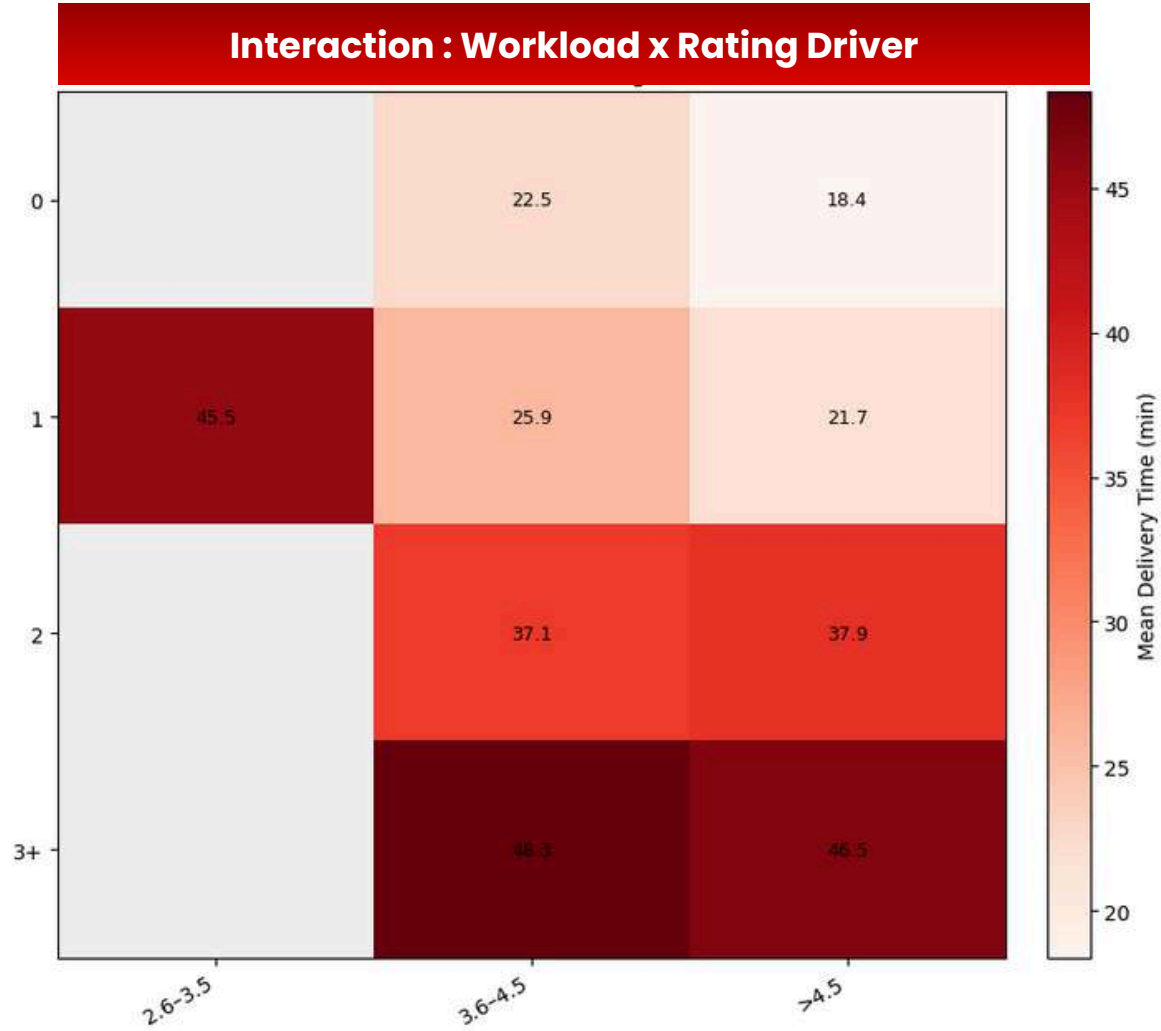
Better vehicle conditions (1–2) reduce delivery times in all weather scenarios, though delays remain noticeable during adverse weather like sandstorms and storms



Interaction analysis



High traffic amplifies delays from workload—delivery time jumps from 24.0 min (0 workload) to 47.3 min (3+ workload). Even under low traffic, high workload (3+) still significantly increases delivery time (33.5 min)



Delivery time spikes dramatically when workload reaches 3+ orders, regardless of driver rating. The fastest times occur at low workload (0 orders) with high ratings (18.4 min)

Interestingly, even when a worker or service maintains a high rating, a consistently heavy workload can still result in significant delays



Tools



Power BI



Streamlit

This visualizations help business decision-making

Show the main delay factors

Show the main delay factors (traffic congestion, bad weather, festivals, high workload) that increase delivery time

Reveal interaction patterns

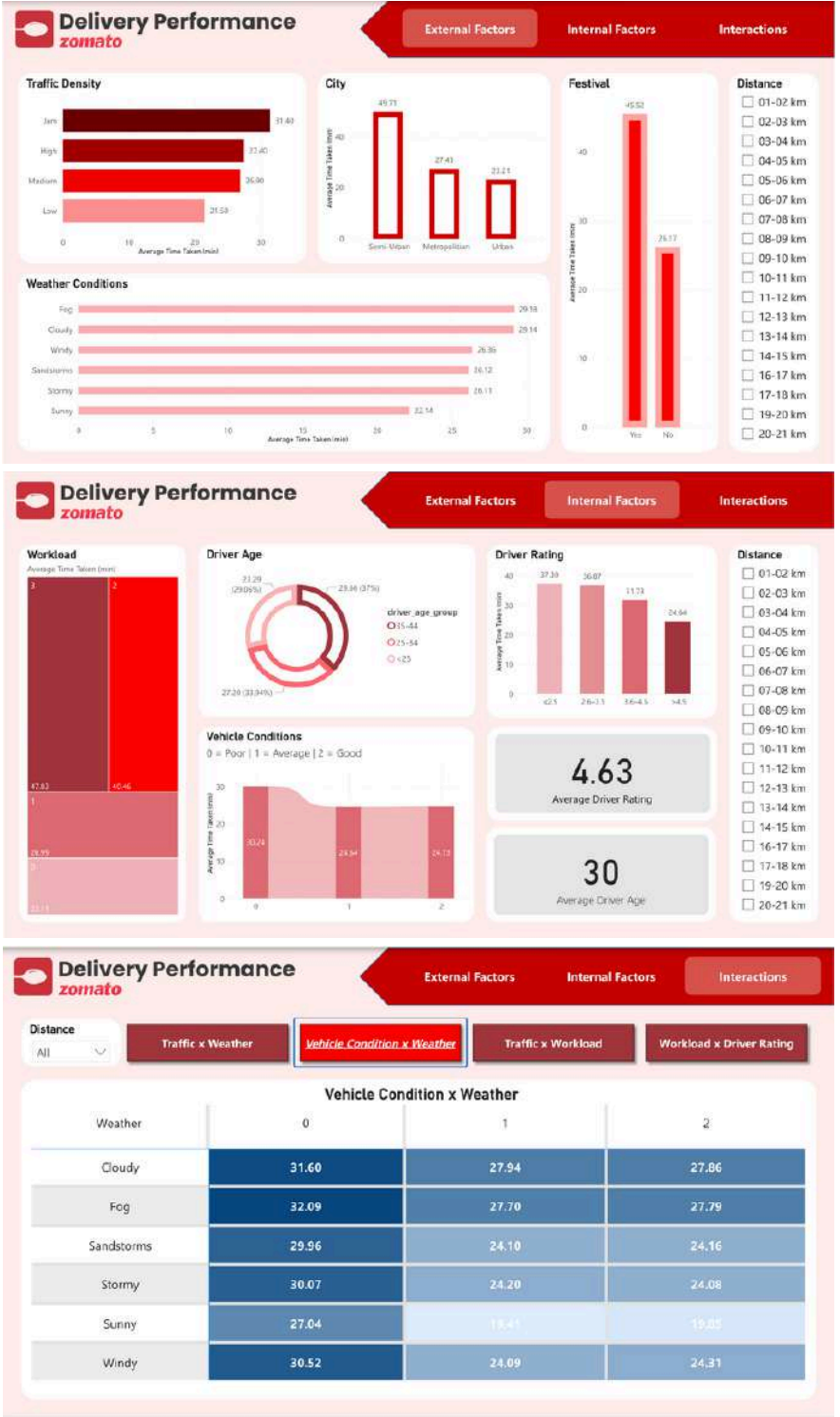
Reveal interaction patterns (e.g., congestion combined with fog results in the longest delays), providing deeper insights that help management design smarter and more efficient route planning to minimize disruptions

Compare performance across regions and conditions

Compare performance across regions and conditions (urban vs. semi-urban, driver rating, vehicle condition) so management can identify where improvements are needed

Dashboard : <http://bit.ly/4fPY5K6>

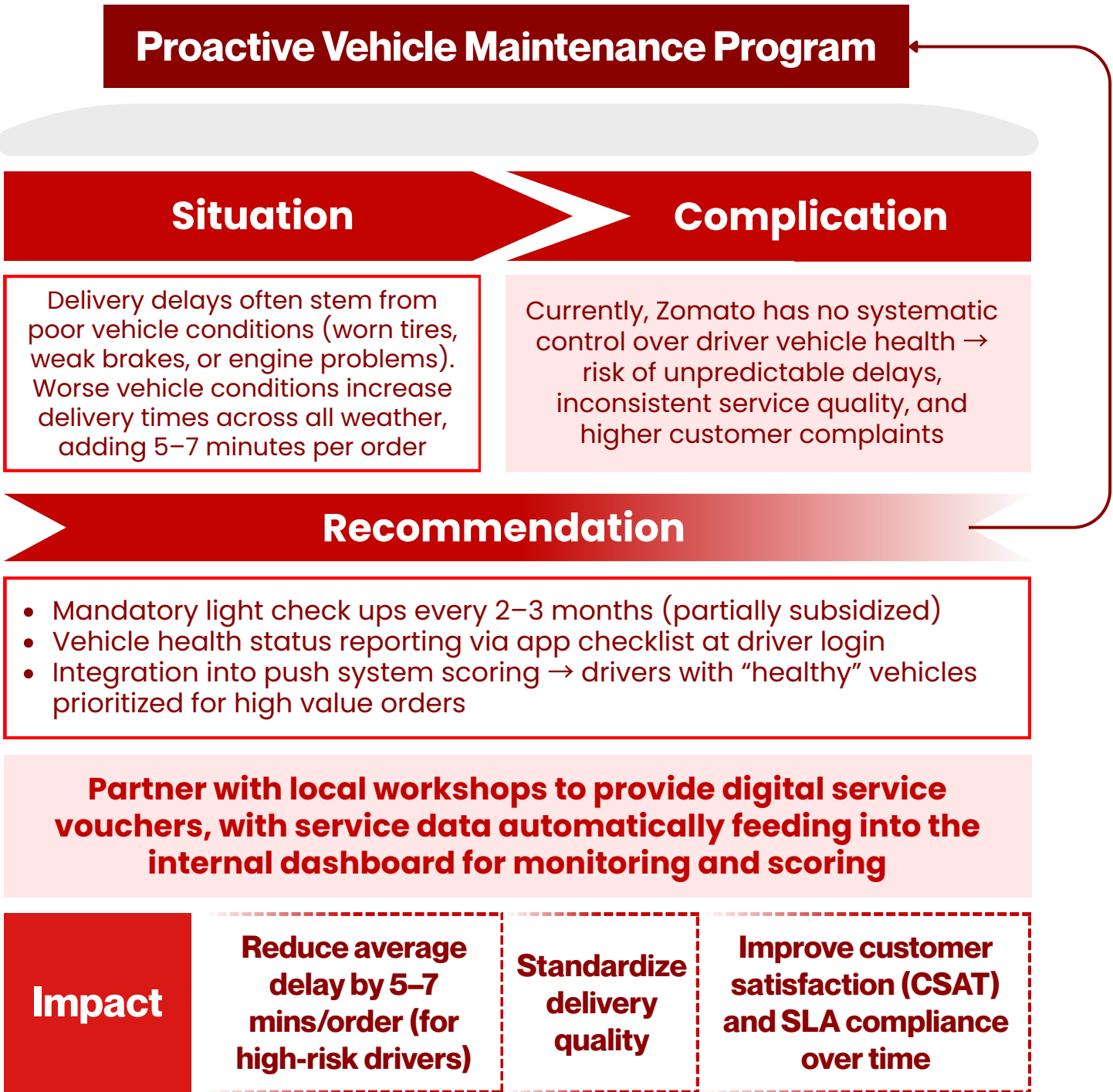
Streamlit : <http://bit.ly/4mqXJfr>



Recommendations & Actionable Insights



**Focus on internal factors (workload, vehicle condition, driver rating) since they are directly controllable by the company. External factors (traffic, weather, festivals) cannot be controlled, but they still need to be anticipated by adjusting internal strategies*



Dynamic Order Allocation



Problem



Data shows significant delays occur when a single driver is assigned 3+ parallel orders. The current push system mostly considers proximity, ignoring other real-time variables

Recommendation

Integrate a dynamic allocation algorithm that accounts for :

Live traffic conditions
(Google Maps API / internal mobility data)

Flexible driver capacity limits
(flexible based on conditions, not fixed)

Historical driver performance
(SLA compliance, ratings, age, etc)

Vehicle condition status
(updated through app or driver checklist)

Implement a priority scoring model → each order is scored and pushed to the driver with the highest compatibility score, not just the nearest one

Impact



Reduce 15–20% of delay cases caused by overloaded drivers



Boost operational efficiency without expanding fleet size



Improve customer satisfaction (faster deliveries) + driver experience (fairer allocation)



Thank you!



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